

The Impact of Social Features and Application Design on User Behavior and Long-Term Engagement of Strava Users

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Abstract

The growth of the health and fitness industry, simultaneously with the emergence of platform-based technology, has led to the incorporation of social features into fitness applications. This study uses the UTAUT2 framework as a theoretical lens to explore the ways in which Strava users interact with the social features embedded into the user experience, as well as how application design affects user engagement. Sixteen semi-structured interviews were conducted to obtain user insights into why and how application design and social features drive user behavior on Strava. Thematic analysis was used to analyze the data, and the findings show the role of application design in driving sustained long-term use. Furthermore, this research offers an adapted version of the UTAUT2 model that better suits the study of platform-based technology use in consumer contexts. This study also highlights the role of social sharing in positively impacting user motivation.

Keywords: health data, fitness applications, social features, application design, UTAUT2

1. Introduction

In recent years, the health and fitness technology market has experienced rapid growth, accompanied by the increased adoption of fitness applications and wearable devices. These apps have evolved from simple activity trackers into interactive platforms that incorporate social and competitive features and personal health management to enhance the user experience (Zhang & Liu, 2024). Current trends reflect a broader shift in which social features have become a core component of digital fitness tools. This blurs the line between social networking and personal fitness tracking, whereby integrating social features into fitness apps holds the promise of boosting user engagement and long-term retention. This study focuses on Strava, a fitness tracking app that has built a global network of over 100 million users and a strong

local uptake in Country A, offering a venue for both international comparison and local context. Strava users share their exercise activities, exchange feedback with peers, and strive for social connection as part of a vast online community (Strava, 2024), positioning Strava as an ideal platform for examining the social dynamics of fitness applications. Prior studies and industry analyses suggest that social features can motivate individuals to be more active and stay committed to using the application (Friel et al., 2019; Teng & Bao, 2022). Tong and Laranjo's (2018) study on Strava found that besides self-tracking motives, social motives such as receiving support or comparing performances with others directly contribute to habitual application use.

The previous literature also highlights that motivational outcomes depend heavily on user characteristics, cultural context, and the specific design of the app's social features. Features that enable social comparison and recognition can simultaneously inspire motivation and/or trigger negative emotions such as anxiety and unhealthy or compulsive behavior, particularly among vulnerable users (cf. Friel et al., 2019; Tong & Laranjo, 2018). Moreover, persistent concerns about data privacy and transparency further complicate the user experience (e.g., Cai & Li, 2024).

These contradictory findings underline the need for a deeper, more nuanced understanding of the ways in which social features in fitness applications influence user behavior. Additionally, there is a gap in the literature on qualitative insights explaining why and in what ways social features influence user behavior and long-term engagement; this gap hinders understanding of the intricacies of social dynamics and how negative user experiences may lead to disengagement or build harmful tendencies for certain user segments. Moreover, there is a lack of research as to the ways in which users perceive the benefits and drawbacks of features in their day-to-day use-experience.

This study, which aims to fill these gaps, is guided by the following research questions: 1) *How do social features in fitness applications influence users' motivation and engagement in physical activity?* and

2) *How do the design and functionality of features in fitness applications enhance user engagement and long-term retention?* Our study contributes especially to the knowledge of Strava's social features, in particular, how feature design shapes user behavior and engagement levels. Moreover, our findings build upon the prior framework of the Unified Theory of Acceptance and Use of Technology² (hereafter UTAUT2) by incorporating social interaction factors into the model to understand fitness application usage. Practically, this research can guide fitness app designers and businesses working with health data in crafting more effective social features that support users' fitness goals. By identifying what aspects of social interaction drive positive behavior and what do not, our research can inform design decisions that improve user retention and satisfaction. By providing a rich qualitative understanding of user experiences with social features, our research extends existing UTAUT2 applications to the domain of socialized fitness tracking.

The remainder of the paper is organized as follows: the next section presents a theoretical background on prior research on fitness applications and our theoretical framework, the UTAUT2 model. The paper then outlines the study's methodology, followed by its findings, a discussion, and conclusions.

2. Theoretical background

2.1. Prior research on users' motivation to use fitness applications

Prior research has, for example, used surveys to study the interrelations between physical activity levels and fitness application use (e.g., Barkley et al., 2020; Teng & Bao, 2022). However, these quantitative studies do not explain why users behave in certain ways or how they perceive the phenomena within the applications. Faizah et al. (2024) took a qualitative approach and used thematic analysis to explain user motivations. The Stimulus–Organism–Response Model (S–O–R) has also been employed to interpret and explain user behaviors, motivational dynamics, and engagement patterns observed in the context of fitness applications (e.g., Teng & Bao, 2022).

Research shows that fitness applications have a positive impact on people's physical activity behaviors because people who use a fitness application are more physically active than those who do not (Barkley et al., 2020). Faizah et al.'s (2024) findings also reflect that fitness applications and their features have a significant influence on physical activity

behaviors and intentions to continue app usage. Faizah et al.'s (2024) research divided the feature types into four categories: self-monitoring, social support, platform rewards, and external influence. Their results indicated that self-monitoring was the most influential feature, while external influence and platform rewards also had a significant influence. However, social support did not indicate an effect on the perceived usefulness of a fitness app.

This view is contradicted by Whelan et al. (2021), whose findings indicate that fitness application use and exercise have a positive effect on exercise habits when motivated by reciprocation and a group of similar-minded individuals. Earlier research (e.g., Whelan et al., 2021) indicates that sharing activities on social media has a positive effect on the sustained use of fitness devices at least partially due to the ability to enhance social support and reinforcement. A study on a gamified exercise service found that people do indeed “work out for likes,” as social approval and reciprocity from others increased users' willingness to exercise and continue using an application (Hamari & Koivisto, 2015). In addition, the sense of belonging to a group of likeminded peers has been found to have a positive effect on the sustained use of and engagement with a fitness application (Liu et al., 2024; Teng & Bao, 2022).

Research on the social features of fitness applications has seen a wide range of studies in recent years. Although many studies highlight the positive effects of social features on consumer behavior (e.g., Friel et al., 2019; Teng & Bao, 2022), the literature also acknowledges that certain users may birth tendencies that can be described as obsessive (Whelan et al., 2021). Moreover, social features that promote recognition—meaning that only positive data and photos of exercise are shared—may lead to harmful perceptions of exercise and burnout over a longer time span. Hu et al. (2023) also argued that overly beneficial perceptions of social engagement on fitness apps are dangerously misleading, especially for those already at high risk of emotional exhaustion. One such group comprises users with poor performance or body image, who may be more likely to suffer from negative feelings such as anxiety and disappointment when exposed to social comparison in fitness applications (Chen et al., 2024). Social fitness applications have also raised concerns regarding data privacy, as applications have been identified as communicating with unexpected third parties (Kazlouski et al., 2020).

2.2. The Unified Theory of Acceptance and Use of Technology 2 framework in a nutshell

The original UTAUT model, which was based on a review and synthesis of eight theories/models of technology use, was developed by Venkatesh et al. in 2003 (Venkatesh et al., 2012). While the initial model was developed for use primarily in organizational contexts, the UTAUT2 model was extended to be more fitting to consumer use, adding to its generalizability. And while the first model was composed of four factors—performance expectancy, effort expectancy, social influence, and facilitating conditions—the second model was extended to include the factors hedonic motivation, price value, and habit (Venkatesh et al., 2012). The relationships between factors and use behavior, as well as behavioral intention, are moderated by age, gender, and experience, dropping the first model's voluntariness.

In their extensive literature review, Tamilmani et al. (2021) categorized research utilizing UTAUT2 into four different categories: general citations, UTAUT2 application, UTAUT2 integration, and UTAUT2 extension. With a wide array of research either integrating (combining with another framework) or extending (including and/or excluding factors), adaptations of the model have been widely accepted within the IS research community and have been used to study many different phenomena.

2.3 Research on fitness applications and wearable technology using the UTAUT2 model

Within the context of fitness and mobile health applications, prior research has focused on using the UTAUT2 framework to study what motivates continued use of health and fitness applications (Yuan et al., 2015). Conducted on college-aged students in the Midwestern United States, their study found that performance expectancy, hedonic motivations, price value, and habit were significant predictors of sustained use, while effort expectancy, social influence, and facilitating conditions did not appear to be significant predictors. However, their findings are partly contradicted by those of Schomakers et al. (2022), who found that social influence influenced the acceptance of health applications in Germany.

Interestingly, most of the prominent research within the field was conducted with questionnaires and used the UTAUT2 framework to predict which factors play a part in behavioral intention and intention to use. Moreover, to our knowledge, the UTAUT2 framework has not been utilized in the context of a qualitative interview research method within the research field of wearable technology and fitness and health applications.

The lack of research utilizing the UTAUT2 framework (Venkatesh et al., 2012) in the interview context to study the research field of health and fitness applications further clarifies and validates the findings of our study. Moreover, we decided to use the UTAUT2 model as a theoretical framework for our study for a number of reasons. First, the UTAUT framework has helped with the distillation of both the critical factors and contingencies relating to the prediction of behavioral intention to use a technology and technology use (Venkatesh et al., 2012). Second, in comparison to UTAUT, the second model focuses primarily on consumer use. Third, the model's adaptations of UTAUT core constructs, such as performance expectancy and social influence, helped us study whether fitness applications provide benefits to consumers performing certain activities and how social circles affect the use of said applications (Venkatesh et al., 2012). Fourth, the additions of hedonic motivation and habit to the original framework helped us dive into concepts of fun or pleasure derived from fitness application use, as well as whether the use has become automatic and a part of fitness itself.

3. Methodology

3.1 Research design and approach

We adopted a qualitative research design (Eriksson & Kovalainen, 2008) to investigate the ways in which the social features of fitness applications impact user behavior, motivation, and long-term engagement with the Strava application. Key constructs from the UTAUT2 model, such as performance expectancy, social influence, hedonic motivation, and habit, were used to inform the development of the interview guide and to analyze emergent themes. Constructs such as facilitating conditions and effort expectancy were excluded from the theoretical framework because the target user group consisted of digitally literate individuals who already possessed the necessary resources, knowledge, and skills to use fitness-tracking applications. Prior research also suggests that these constructs tend to show limited variance and reduced explanatory power in contexts where technology is widely adopted and users are proficient (Venkatesh et al., 2012). Therefore, to streamline our research and focus on the more impactful determinants of usage behavior, these variables (facilitating conditions and effort expectancy) were left outside the scope of this research. Finally, we added an additional construct, platform switching propensity, to gain insights into

why and how users adopt or might move on from Strava.

3.2 Data collection and ethical considerations

The data for this study were collected through semi-structured interviews guided by the UTAUT2 framework. Semi-structured interviews provide a set of themes for discussion, but they still allow a variety of question types, including exploratory “what” and “how” questions, to surface information (Eriksson & Kovalainen, 2008). Additionally, interviews as the data collection method enabled us to gain direct insights into users’ actual experiences with Strava use. Participants consisted of individuals with varying experience levels of both activity experience and Strava use, which allowed for a more in-depth understanding of the app’s social dynamics and long-term user experience. Participants were selected based on their active use of the Strava platform, ensuring that they had direct experience with the app’s social features. Participants were selected to represent young adults in the 25–33 age range from the capital city area of Country A. The selected sample size comprised 16 participants, which helped to reach theoretical saturation while remaining manageable for analysis. Interviewees were contacted through social media and direct messaging channels. The interviewed individuals’ background information is presented in Table 1.

Interview- wee #	A ge	Gender	Activity level per week	Years of Strava use
1	26	Male	4–5 times	3 months
2	26	Female	3–4 times	1.5 years
3	25	Female	2–3 times	1 year
4	25	Female	3–4 times	1.5 years
5	25	Female	5 times	4 years
6	27	Female	2–5 times	3 years
7	26	Male	5 times	1 year
8	26	Male	8 times	1 year
9	28	Male	4–5 times	3 years
10	28	Male	5–6 times	1 month
11	25	Female	3 times	1.5 years
12	28	Male	1–4 times	4 years
13	26	Male	5–6 times	1 year
14	25	Female	5 times	1 year
15	25	Female	3 times	1 year
16	33	Male	7 times	1 year

Table 1. Interviewee background data

Interviews were conducted either in person or via video conferencing platforms, depending on the participant’s preference. Each interview lasted from 30–60 minutes. With the participants’ consent, all interviews were audio-recorded for subsequent transcription and analysis. We also used the mirroring

technique to our advantage when applicable to keep the focus on the interviewees’ language (Myers & Newman, 2007). We have carefully translated relevant quotes from Country A’s language into English.

Prior to conducting the interviews, we obtained explicit permission from participants to record the sessions. We also made it clear that participation in the study was entirely voluntary, and participants were informed that they could stop the interview at any time without needing to provide a reason. Furthermore, they were not obligated to answer any questions that made them uncomfortable, such as gender. We offered each interviewee the chance to receive the guiding questions for the interview beforehand. Moreover, prior to the interviews, we explicitly stated that anything the interviewee wanted redacted could be left out of the research, and that we would always respect each request without question. This also highlights our commitment to ethical research practices and to fostering a space where participants felt in control of what they shared and how it would be used. To ensure data security and confidentiality, all interview recordings, transcripts, and notes were stored securely within the authors’ organization’s Microsoft environment, which complies with institutional data protection policies. All identifiable information was removed or anonymized during transcription to protect participants’ privacy.

3.3 Data analysis

While the interviews were guided by the UTAUT2 framework, the data were analyzed inductively using Braun and Clarke’s (2006) thematic approach, which allows themes to emerge naturally. The authors independently coded the transcripts and then cross-checked their results. Braun and Clarke’s approach to thematic analysis involves six phases, helping to ensure a rigorous analysis process. Although we adhered fully to all six phases in our analysis, for clarity and succinctness, we later consolidated their original phases into three broader stages.

First, we searched for common patterns to identify themes. This involved grouping related interview answers together to form broader themes that encapsulated significant aspects of the participants’ experiences with Strava. Each theme was carefully considered to ensure that it accurately reflected the underlying data and was relevant to the research questions. The second step involved checking each theme against the interview answers to ensure that they accurately represented the data. During this process, some themes were merged or split based on their relevance and the richness of the data they

encompassed. In the third step, we conducted a detailed analysis to define what each theme represented. The focus was on articulating how each theme contributed to understanding the social dynamics and design features of Strava and their influence on user behavior. Throughout the analysis, we remained reflexive and attentive to our biases, striving to let themes emerge from the data rather than imposing our expectations.

4. Findings

In this section, we present the key themes from the semi-structured interviews. We then use the UTAUT2 framework holistically to examine how these themes relate to app use.

4.1 Social comparison, connection, and eligibility

Many of the interviewees mentioned that the combination of roles (the creator of a posting and the consumer of a posting made by someone else) on Strava creates what can be described as a “social mirror.” Interviewees mentioned that seeing the work put in by others motivated them to exercise. As described by Interviewee 16, *“In my case, seeing the friends that I usually train with doing super hard work, seeing their weekly mileage going up, makes me not want to disappoint them in a sense. So yes, I see it as something that motivates me, and I want to keep up the effort for the group that is training together.”*

Moreover, being motivated by other users’ shared experiences can also provide inspiration for a specific exercise or route. Interviewee 2 shared, *“Seeing that other people have exercised boosts my own motivation to train. Sometimes, when I feel a bit anxious about not having trained, I can see that someone has run a really fun route or a longer distance, making me want to do a similar exercise myself. [...]”*

Knowing that friends would be able to see their performance created a sense of accountability. Several interviewees highlighted that sharing personal exercise data also worked as a way to reinforce commitment to exercise habits and encouraged them to perform better during their sessions, meaning they would put more effort into the training so that the data shared later would look better to their followers. Interviewee 9 stated, *“It [sharing workouts] is something that I may think about during exercise, where I might run a little faster due to that. Sometimes I may also feel a bit better after the workout when I see that the pace is a bit faster [...]”*

Some interviewees felt pressure from the continuous comparison that is present on the platform, which led to feelings of inadequacy about their own workout routines or exercises. Interviewee 11 said, *“I definitely felt more pressure about sharing my workouts in the beginning of my running journey. I felt that I needed to run faster or longer and do my best in a workout so that it looks OK.”*

Three interviewees (6, 9, 11) said that while they experienced pressure from comparison to have a positive impact, they could also see some negative sides to it. For example, Interviewee 6 stated, *“I feel that my Strava bubble has become quite wide and polarized, which I feel has a negative impact on me. As the application has become almost like social media, I feel pressure to post something, and when I don’t necessarily have time to exercise as hard as I want, it has a negative impact on me.”* Overall, the interviews suggest that the Strava features that enable social comparison play a multifaceted and complex role in users’ engagement and exercise habits. Being recognized by other users on posted exercises is a core strength of Strava. While virtual affirmation through likes or “kudos” on Strava played a crucial role for some of the interviewed users and boosted their hedonic motivation and habit, others did not place much emphasis on it. Some even decided to refrain completely from using the feature.

For some users, receiving kudos and comments on posted exercises worked as a powerful external motivator by providing immediate social reinforcement. Interviewee 15 stated, *“I get similar dopamine from the social interactions on Strava as I would using other social media applications. I visit the application even when I am not posting any exercises just to interact with other users. I would think that it is most likely the reason that I have been using the application as long as I have.”*

Other users mentioned that while they did not rely heavily on social feedback, occasional interactions via kudos and comments provided a welcome boost in validation. Together with intrinsic motivation, acknowledgments from others played a supportive role in confirming individual progress. Interviewee 4 stated, *“[...] of course it is a fun aspect related to using the application. And in some sense, it does motivate me when I get kudos, but it isn’t necessarily the main driver of why I use the application. I have turned off the notifications on the application as well, so I don’t really visit the application if I am not posting anything.”* This indicates that while not actively using Strava’s social features that enable user interaction negatively affected the users’ habit, it did not diminish their performance expectancy or hedonic motivation.

4.2 Performance tracing and data utilization

Based on the interviews, *training loggers* used Strava as a convenient way to record their activities without delving deeply into the available metrics. For them, tracking workouts was more about maintaining consistency and reflecting on past efforts than using detailed analytics to improve performance. Interviewee 5 explained, *"I just like to see my runs in one place. I don't really analyze them; I just log them and move on."* Conversely, *data-driven athletes* took a more analytical approach, closely examining their pace, heart rate, and prior exercise loads to adjust their training. They actively used Strava's features and collected data to optimize their workouts. For these users, Strava was more than just a log; it informed and guided their training decisions. Moreover, these users could see the additional value of even more analytical features. Interviewee 16 emphasized, *"If I had a tool that functioned as a second coach, analyzing my progress and providing structured guidance, it would significantly impact my training."*

While training loggers may benefit from a simplified, intuitive interface that prioritizes ease of use, data-driven athletes seek deeper insights and personalized feedback. Thus, based on the interviews, the challenge lies in balancing both user expectations, ensuring that casual users are not overwhelmed while still catering to those who rely on detailed analytics.

Many interviewees stated that Strava's goal-setting feature allowed them to set distances or time-based objectives for specific periods, such as monthly running or cycling distance targets. This feature was particularly used by individuals training for races or long-term fitness improvements. Additionally, some stated that the Strava clubs of which they were a part integrated goal setting into their leaderboards, thereby creating an extra layer of social motivation and accountability. While some participants appreciated the structured approach, others noted that they rarely interacted with it. One interviewee mentioned that setting a weekly kilometer goal provided additional motivation, as progress tracking helped maintain consistency: *"Setting weekly distance goals motivates to exercise more. I may find myself in situations where I am looking at my Strava during the week and try to plan that, OK, I need to run this much on that day and this much on that day to reach my weekly goal"* (Interviewee 12). Interviewee 16 stated that they engaged more with club-based goal-setting leaderboards, whereby collective achievements and rankings drove motivation. However, some users remained unaware of this goal-setting functionality or found it irrelevant to their fitness routines. Interviewees 12 and 9 also expressed that tracking

goals could sometimes introduce unnecessary pressure, making exercise feel more like an obligation than an enjoyable activity.

Strava leaderboards serve as a social comparison mechanism, ranking users based on performance in activities such as running and cycling. Based on the interviews, the findings suggest that while leaderboards can act as a strong motivational tool, they may also contribute to pressure and self-comparison. Some users reported feeling inspired when they saw their ranking improve, particularly when competing within smaller peer groups. However, others found that leaderboards were more relevant for elite athletes than for casual users, who focused on personal improvement rather than competition. For some individuals, friendly competition introduced a healthy level of motivation through gamified data: *"I enjoy segments and the leaderboards, especially when cycling. Once I got a notification that someone had passed me in a challenge, and on my next ride I deliberately chose that route to try to win the segment back"* (Interviewee 5). In contrast, Interviewee 14 felt disengaged from leaderboards, stating that their primary focus was on tracking personal progress rather than comparison with others, highlighting the differences in user preferences toward such features.

4.3 Self-presentation: Strava is more than a fitness tracker for some users

Many interviewees stated that they did not necessarily decide not to post certain exercises on the platform due to negative performance but because they did not want to overcrowd someone else's feed with "unnecessary" exercise posts. Similar feelings were shared by other interviewees (2, 8, 13, 7, & 9), showing that public perception played a role in what was seen as acceptable to share and thus highlighting the role of social influence in the use of the platform. Interviewee 2 shared, *"Sometimes I decide not to share exercises that I think others do not want to see. Sometimes if the activity is shorter, it feels wrong to share it; I find myself wondering if this is even a workout and think about what others would think about it."* Interestingly, while most interviewees shared the same notion, only Interviewee 2 stated that they had been directly told not to post certain types of activities on the application. While others tried to curate their self-image on the application to depict a more serious athlete, other interviewees consciously tried to battle the phenomenon. Interviewee 6 shared, *"I have always had the principle that I try to post all my workouts no matter how small or slow or easy they are to keep as real a picture as possible. I feel the main reason for the application is to give encouragement to*

others and not to flaunt with the most impressive achievements."

Most interviewees who used the application to track their personal progress stated that the ability to see their progress and prior workouts helped them build a narrative of how they had evolved in their chosen activity. Interviewee 15 stated, *"I'd say that prior data is a big reason why I use the application and [...] I'll be using the application for a long time. For example, the application just produced a summary of my last year, where I could see what I had done and how many days I had trained."* Interviewee 16 chose the application because they needed to have a historical record of what they had previously done. In addition to keeping users habitually engaged while increasing hedonic motivation and helping them plan future workouts, building narratives through individual progress tracking increased hedonic motivation by showing proof of development: *"Seeing visuals of how I trained in a given month and being able to compare it to current activities definitely gives me motivation"* (Interviewee 13).

Many interviewees found that the ability to share their activities through images and descriptive texts transformed the application from a performance tracker to a social fitness community. Interviewee 5 shared, *"For me, the use of Strava has become almost like social media. Nowadays when I post activities, I almost always add pictures and things like that."* For some users, the features allowed them to contextualize and add depth and storytelling elements to their training. Interviewee 15 shared that they liked to give meaningful titles and descriptions to their workouts, thereby shaping how others saw them: *"I like the possibility of adding pictures, titles, and descriptions to my workouts. I think that this is a nice combination of the training itself, but also sharing how I felt that day or who I was training with and how enjoyable it was."* Interviewee 6 also saw the shift from traditional tracking toward a more social space, although not everyone felt completely positive about the change: *"It [Strava] has become more and more like social media, which I am not the biggest fan of. But in some sense, as everyone is making more visually compelling posts, it has some qualities to it that I enjoy"* (Interviewee 6).

Many interviewees stated that while they did not actively add descriptive components to their activities, they found enjoyment from other users' posts. This suggests that even passive users derive sustained engagement and motivation from viewing narratives built by others. At the same time, the ability to shape their workouts into stories through descriptions, images, and reflections adds a more engaging and enjoyable dimension to the experience. Seeing others

share their training and achievements influences behavior, whether by providing inspiration, fostering a sense of accountability or strengthening the connection to a broader fitness community. The combination of these elements makes Strava a place where both personal progression and social storytelling merge, enforcing users' habits and increasing hedonic motivation.

4.4 Habit integration and routine description

Interviewees described how their use of Strava's social features had evolved over time and how the app fit into their long-term fitness routines. For many, uploading an activity to Strava had become a natural part of working out. Participants reported that they usually opened Strava (or let their device sync to Strava) whenever they finished a run, ride, or other tracked workout. Some even mentioned that they checked the app daily, similar to other social media applications, even if they did not post every day. The habitual integration of Strava usage into workout routines was identified as having different levels depending on the user and how meticulously they approached training in general.

The interviews show that, for more serious and goal-driven athletes, meticulously recording every detail on Strava can feel essential, whereby the habit enables more efficient progress tracking. We found that a common pattern in long-term habitual use was an initial phase of high engagement with social features, followed by tapering to steadier, more selective usage. Many interviewees stated that they shared more information, wrote more detailed captions, and actively followed others early in their application-use experience. Interviewee 15 reflected, *"sharing in more detail was fun in the beginning when everything was new. [...]"*

Interviewee 6 stated that while they initially spent a lot of time learning and adjusting to the application, using the application became easier over time. Some interviewees also experienced seasonal or contextual fluctuations in use. Possible breaks into habits were caused by a lack of interest in exercise in general, external factors such as injuries, and weather-related reasons. Interviewee 8 stated, *"I typically use Strava about 1–2 times a week. However, I haven't been using it lately because the weather has been so bad that I have not been running."* In general, users may develop usage habits for different reasons, emphasizing the application's design process to create the ability to cater to many different users.

4.5 Perceived value of free vs. premium features and subscription behavior

Most participants expressed satisfaction with the free version and uncertainty about what additional value the premium subscription offered. Many users did not clearly know which features were exclusive to premium: *"I actually don't even know what's included in premium. Maybe it adds value, maybe it doesn't, it's hard to tell"* (Interviewee 5). Interestingly, several participants only discovered that certain features (such as goal setting or route planning) were behind a paywall when it came up in the interviews.

One reason for choosing the premium feature was the route planner and related map functions. Interviewees valued being able to create and sync routes to their devices, which was a reason to pay for at least part of the year due to the seasonality of Country A: *"I'm using the free version now, but I had premium for a couple months because I needed the feature to draw my own routes. But in the end, I didn't even use it that much, so I switched back to the free version"* (Interviewee 8).

Interviewee 13 mentioned advanced metrics such as Grade Adjusted Pace as a niche premium feature they used, although they admitted they had not explored all the premium functions. Only one interviewee, a current subscriber, felt they got a lot of value for their money with premium, praising its deeper analysis and considering the price reasonable. More commonly, interviewees said they would consider subscribing only if certain conditions were met, such as if the user became more serious about training and needed detailed performance data or if Strava moved a currently free, beloved feature behind a paywall. A few noted that if a free feature they relied on, such as basic activity statistics or the social feed, suddenly required premium, they might feel "forced to pay" out of fear of missing out (FOMO) or necessity.

4.6 Privacy concerns, barriers to use, and possible switching factors

Some users expressed caution regarding sharing their exact location, especially around their home addresses and frequently visited areas. Several other interviewees also reported using Strava's privacy settings, such as hiding the start and end points of routes, to reduce potential risks: *"I have taken to using Strava's hide start and end points function to hide my exact home address, just in case"* (Interviewee 3). While most participants acknowledged that Strava collects and stores health-related information, they either felt that they had no real alternative if they wanted to use the platform or that the data were unlikely to be misused. Interviewee 1 stated, *"I am concerned that the application gets all my personal*

data, and I can't really know what they do with it. [...] I think it is quite worrying that such applications have access to so much personal data of users."

In addition to privacy, some participants experienced inaccurate GPS tracking, particularly in areas with poor satellite reception (which affected their workout data), as minor barriers to use. Battery drain during long activities and occasional syncing problems between Strava and third-party devices were also mentioned. Even the occasional inability to instantly upload and view activities may reduce the sense of routine and weaken engagement over time. However, these technical limitations were not seen as critical enough to deter users from continuing to use the platform.

Many interviewees (7, 14 & 15) highlighted that a direct recommendation from a peer using the application was their initial reason for joining the application. Other reasons for joining were related to the application's features, such as route planning, whereas others found the interface of the application superior to the application they were using previously. Interviewee 10 stated, *"Once I began running again, I switched over to Strava, mostly for the reason that it was easier to use than the application I was using before [...]"*. Interestingly, Interviewee 3 said that she was "forced" to make the switch once they acquired a new wearable tracking device that did not support the connection to the old application: *"[...] I received a new watch as a gift and it did not connect to the application, so I was forced to start using Strava instead. I lost all my years of data in the switch, which I was initially really annoyed about"* (Interviewee 3).

Most interviewees stated that the strongest reason for switching to another platform would be if their friends and peers moved to an alternative application. Some interviewees highlighted that they would switch to another application if it offered superior features (e.g., AI training insights).

5. Discussion

This qualitative research aimed to answer the following research questions: 1) *How do social features in fitness applications influence users' motivation and engagement in physical activity?* and 2) *How do the design and functionality of features in fitness applications enhance user engagement and long-term retention?* This study used an adapted version of UTAUT2, both as a theoretical framework to guide the data collection and as a holistic lens through which to connect the emerged themes to fitness application use.

While the original framework by Venkatesh et al. (2012) was intended for researching phenomena

related to new technology use in the consumer context, we argue that its original composition is not fully applicable to understanding new technologies, specifically platform-based applications, and their use. To accommodate this, we adapted the UTAUT2 model by dropping the constructs facilitating conditions and effort expectancy because they offer little additional value in contexts where the technology is widely adopted and users are proficient in its use (Venkatesh et al., 2012).

Given the network effects in place and the concept of platform-swinging—where users tend to strategically move between platforms (Wang et al., 2024)—we incorporated the construct of Platform Switching Propensity (PSP) into the UTAUT2 model. This addition enabled us to better understand users' willingness or tendency to move from one application to another due to dissatisfaction, superior alternatives, or changing social circles, reflecting the fragility of platform loyalty in consumer app ecosystems.

Our findings support previous research (Friel et al., 2019; Whelan et al., 2021), as interviewees stated that sharing exercise activities with peers positively impacted their motivation to exercise as well as their intention to keep using the application. This finding contradicts those of Faizah et al. (2024), who stated that social support did not have a significant impact on the perceived usefulness of an application. Furthermore, interviewees stated that they thought about post-exercise recognition during their actual workout, which most participants agreed provided motivation (cf. Hamari & Koivisto, 2015). However, our findings explain that positive motivation was not only limited to how hard a user may exercise, but also factored in the quality of exercising and how to exercise.

Although most interviewees agreed that engaging with Strava's social features had a positive effect on both their motivations to use the application and to exercise, our findings also indicate the reasoning behind negative trends found by, for example, Chen et al. (2024), whereby users tend to curate which activities they decide to show their peers. Our findings indicate that most of the reasons behind selective sharing were reflections on what others would see as "real" or "actual" exercises. The trend of maintaining a positive social image, which is not always healthy, can also happen indirectly in situations where users curate their shared posts with positive intentions. This view is very much in line with Hu et al. (2023), who highlighted the dangers of overly positive depictions of social interactions within fitness applications.

Most users in our study indicated that their initial adoption of Strava was driven by peer influence. Many users had also previously relied on alternative fitness

applications, such as Sports Tracker or Nike Run Club, and reported switching to Strava primarily to remain connected to their social network. This view builds on existing research by Liu et al. (2024), whereby group-level social interaction designs were seen as valuable elements for fitness applications.

Our findings also challenge the view of Liu et al. (2024), arguing that application design should focus primarily on creating social interactions among users and not necessarily drive grouping mechanisms. Interestingly, our research also identified outliers who experienced the opposite effect of increased social connectivity. Rather than enhancing engagement, a growing number of connections led these users to become more selective about what they chose to share. Some participants noted that having a large network on Strava resulted in overly cluttered social feeds, which diminished the quality of their user experience and reduced their motivation to post and browse activities. Our findings also offer valuable insights into why and how users value certain types of features, building on prior research by Faizah et al. (2024) and Hu et al. (2023), which has highlighted which features are seen as valuable.

Our findings highlight that, while feature abundance may be crucial for catering to a wide array of users, individual users may tend to have a more "quality over quantity" approach, valuing certain features that fit their needs. In essence, feature depth provides multiple hooks for different motivations; one user might stay for the community, another for the detailed metrics, and another for the gamified challenges. Importantly, many of these features—such as goal setting, self-monitoring, feedback, and rewards—work together to create a habit. Our findings indicate that the logging of exercises and the social dynamics brought by such applications help create habits for users in which they look forward to application use during or after exercise due to the hedonic motivation boost from interactions with peers. Furthermore, when an app consistently engages users on multiple fronts, it fosters a sense of loyalty, in which using the app becomes an ingrained part of a user's health and fitness routine.

6. Conclusion

This qualitative study used the UTAUT2 framework as a theoretical framework to explore how young adult users of Strava interact with the social features embedded into the user experience, as well as how application design affects user engagement. Sixteen semi-structured interviews were conducted to glean user insights into why and how application design and social features drive user behavior on

Strava. The findings of this study offer an adapted version of the UTAUT2 model that better suits the study of platform-based technology use in consumer contexts. This study also highlights the role of social sharing in positively impacting user motivation.

We acknowledge that our study has limitations. First, this study is based on a limited sample of Strava users who may not fully represent the broader population of fitness application users. Since user engagement varies widely across different demographics and this study is limited to one application that heavily integrates social dynamics and competition into its design, the fact that other applications may emphasize other motivational mechanisms has been neglected. Further research could, for example, examine how individual user differences, such as personality traits or cultural differences, mediate engagement with social features. Longitudinal studies could also help better explain how external factors such as seasonal changes affect engagement with fitness applications.

7. References

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